

VEHIGUARD

An Intelligent IoT-Based Multi-Sensor Vehicle Security, Geofencing & Accident Detection System

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Project Guide-

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Problem Statement & Objectives

Problem Statement

Traditional vehicle tracking systems mainly provide **GPS location**, but cannot fully determine what is happening to the vehicle.

Major Problems

- Silent towing when ignition is OFF
- Unauthorized door/boot opening
- Physical tampering and vibration
- Possible accident/impact
- Geofence violations
- Overspeeding
- False alarms from single sensors

Objectives

- Integrate **GPS + IMU + vibration + door + ignition**
- Detect **theft, towing and intrusion**
- Detect **possible accident/impact**
- Implement **geofencing and overspeed monitoring**
- Use **Rule Engine + ML**
- Provide **real-time web dashboard**
- Store historical vehicle events

Literature Review I

Research Papers — 2022–2024

Paper	Year	What They Are Doing	Research Gap
Road Traffic Crash Detection using Machine Learning	2022	Uses sensor/data-based methods for automatic crash detection	Mainly focuses on accident detection; vehicle security is not integrated
GNSS/IMU-Based Vehicle Localization	2023	Combines GNSS and IMU for improved vehicle positioning	Focuses on localization , not theft/tampering/accident-security integration
GPS Spoofing Attack Detection Framework	2024	Uses ML to identify abnormal/spoofed GPS behavior	Addresses GPS security , but not physical vehicle security
IoT-Based Automatic Vehicle Accident Detection & Visual Reporting	2024	Uses ESP32/GPS and sensors for accident detection and reporting	Focuses on accident reporting , not complete anti-theft monitoring
IoT Accident Detection & Alert System with Location Tracking	2024	Detects accidents and sends GPS-based emergency alerts	Limited integration of theft, door, vibration and ML anomaly detection

Literature Review II

Research Papers — 2025–2026

Paper	Year	What They Are Doing	Research Gap
Smart Vehicle Security System: Real-Time Tracking & Theft Prevention	2025	GPS tracking, IoT monitoring and theft prevention	Does not provide the same multi-sensor correlation + ML anomaly workflow
IoT-Based Vehicle Security System: Real-Time Monitoring & Event Logging	2025	GPS, motion/vibration sensing and web-based monitoring	Security monitoring is provided, but integrated rule + ML decision architecture is limited
IoT Vehicle Accident Detection using ESP32 with GPS Tracking	2025	ESP32 + GPS + accident detection + emergency alerts	Primarily accident-focused; lacks complete theft/tampering correlation
Smart Motorcycle Security & Tracking using RFID and GPS	2026	ESP32 + RFID + NEO-6M GPS + tracking	Mainly authentication/tracking; lacks IMU impact + vibration + door + ML integration
Multimodal Vehicle Security System with GPS	2026	GPS + fingerprint + face recognition + vehicle tracking	Strong authentication, but does not combine our physical multi-sensor + impact + geofence workflow

❏ **Overall Research Gap:** Existing systems address vehicle tracking, accident detection, authentication, or theft detection separately or in limited combinations. VEHIGUARD integrates these functions using multi-sensor correlation and a two-tier Rule + ML architecture.

Proposed System

VEHIGUARD — What We Are Using

Hardware

Component	Purpose
ESP32	Main controller
NEO-6M GPS	Location & speed
MPU6050	Acceleration, tilt & impact
SW-420	Vibration/tampering
Reed Switch	Door/boot status
Ignition Switch	ON/OFF status
Buzzer + LED	Local alert

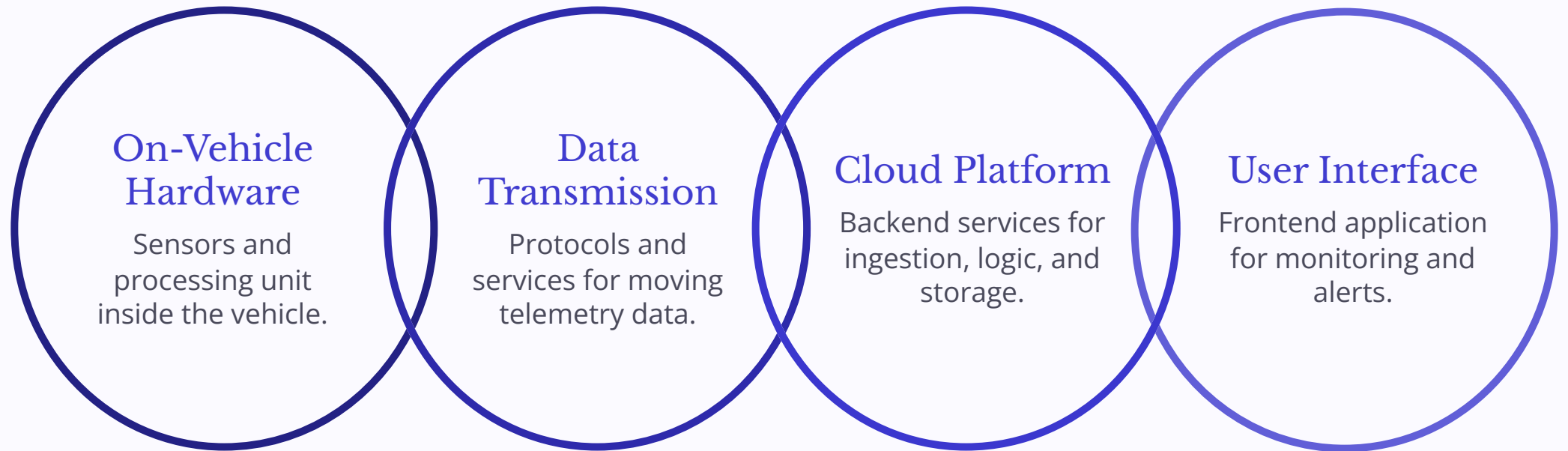
Software

- Arduino IDE / C++
- MQTT / Mosquitto
- Python / FastAPI
- Random Forest / Scikit-learn
- SQLite
- React + Leaflet

These components and their intended functions are specified in your synopsis.

System Architecture

VEHIGUARD Data Flow



Dashboard: Live Location | Speed | Sensor Status | Alerts | Geofence | Event History

Working / Flow

VEHIGUARD Working

01

Sense

GPS + IMU + Vibration + Door + Ignition

02

Preprocess

Calibration + Filtering + Acceleration/Tilt

03

Decision

- **Impact detected?** → Possible Accident Alert
- **ARMED + Door Open?** → Intrusion Alert
- **ARMED + Speed > 5 km/h + Ignition OFF?** → Silent Towing Alert
- **Vibration + Ambiguous Movement?** → Random Forest ML
- **Outside Geofence?** → Geofence Alert
- **Speed > Limit?** → Overspeed Alert

04

Communication

JSON → MQTT → FastAPI

05

Output

Database + Real-Time Dashboard + User Alerts

Our Approach, Contribution & Outcome

How VEHIGUARD Addresses the Gaps

1

Existing Research

Individual solutions: **Tracking** · **Accident Detection** · **Theft Detection** · **Authentication**

2

VEHIGUARD

GPS + MPU6050 + SW-420 + Door + Ignition

3

Multi-Sensor Correlation

4

Two-Tier Decision

Rule Engine: Towing · Intrusion · Impact

Random Forest: Ambiguous/abnormal patterns

5

Complete IoT Pipeline

ESP32 → MQTT → FastAPI → ML/Rules → SQLite → Dashboard

Expected Outcomes

Real-time vehicle tracking

Silent towing detection

Door intrusion detection

Tampering detection

Possible accident alerts

Geofence alerts

Overspeed monitoring

ML anomaly classification

Historical event logs

Real-time web dashboard

VEHIGUARD goes beyond GPS tracking by understanding vehicle status through multi-sensor correlation.